Software Engineering II — 2026-II
Computer Engineering Program, School of Engineering
Universidad Nacional de Colombia

Workshop No. 2 — Design Artifacts and System Modeling

Public Module of the youth football tournament platform: public website, widgets and PWA

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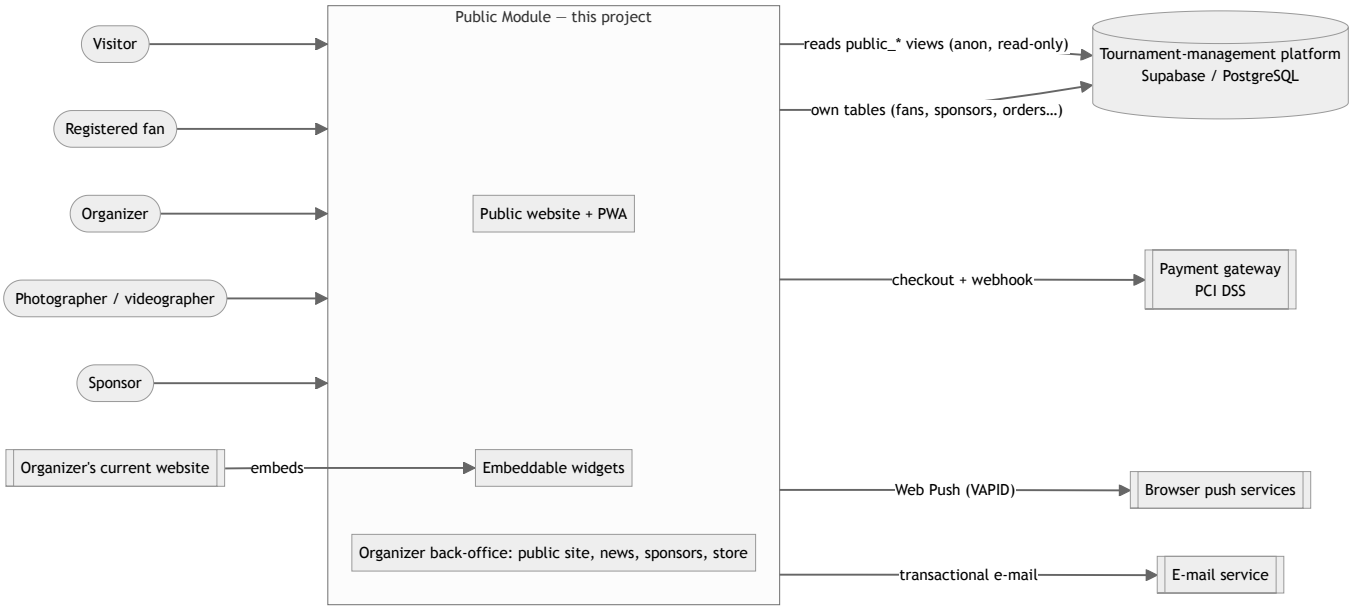
October 15, 2026

1. System domain

1.1 System context

The project extends an existing **tournament-management platform** for youth football (registration, player ID cards, treasury, groups and fixtures, match scheduling, referees, results and standings). That platform is already in production with a real organizer, **DBS Complejo Deportivo**, and stores everything in PostgreSQL (Supabase) with row-level security (RLS) per organization.

This semester the team builds the **Public Module**: the tournament's public website, embeddable widgets and installable app (PWA), plus its commercial features. The Public Module **does not manage the tournament**. It **reads** what the management platform has already approved and published, and it **owns** a new set of data: fan accounts, followed teams, news and galleries, sponsors, the store, photo/video bookings and team pre-registrations.



Boundary rules of the context

Inside the Public Module	Outside (consumed, never modified)
Public pages, PWA, widgets, fan accounts, follows, notifications to fans, news, galleries, sponsor placements and metrics, store, orders, photo/video bookings, pre-registrations, public-site branding	Tournaments, categories, clubs, teams, groups, matches, goals, standings, documents, venues and fields (owned by the management platform)

1.2 Actor characterization

Actor	Type	Description	Main goals in the Public Module	Access
Visitor	Primary, human	Parents, relatives, players and the general public. Mostly on a phone, often at the venue on match day.	Check the schedule, results, standings, top scorers, team and club pages, news and public documents; install the app.	Anonymous, read-only.
Registered fan	Primary, human (specializes Visitor)	A visitor who creates an account with e-mail or Google.	Follow teams, see "my teams" first, receive push notifications, buy in the store, book photo/video packages.	Fan account; only their own data (RLS). It never grants access to the delegate or organizer panels.
Organizer	Primary, human	Staff of the organization (e.g., DBS) who already uses the management platform.	Brand the public site, choose what is published, post news and galleries, generate widgets, manage sponsors, products, orders, bookings and pre-registrations.	Organization member; scoped to their organization.
Photographer / videographer	Primary, human	Contractor of the organization who covers matches.	See the matches assigned to them and upload the delivery link of each booking.	Own account; only their assignments.
Sponsor	Primary, human	Brand that pays for placements on the site, app and matchday graphics.	Check and download the metrics of their own placements.	Own account; only their placements.
Tournament-management platform	Secondary, system	Existing system and single source of truth of the competition.	Publishes approved schedule, results, standings and documents through `public_*` views.	Database views; anon role only sees tournaments marked `public`.
Payment gateway	Secondary, system	PCI DSS certified provider (e.g., ePayco, already integrated in the platform).	Charge store orders and bookings; confirm through a webhook.	Server-to-server, signed webhook.
Push service	Secondary, system	Browser push services (FCM, Apple Push, Mozilla).	Deliver notifications to fans' devices.	Web Push with VAPID keys.
E-mail service	Secondary, system	Transactional e-mail provider.	Order confirmations and delivery links.	API key, server side.

1.3 Business rules

ID	Business rule	Source
BR-01	Only tournaments marked as **public** by their organization are visible in the Public Module; everything else returns no rows.	RF-14, management platform design

ID	Business rule	Source
BR-02	A match is shown on the public schedule only after the organization **approves** the matchday schedule; until then it appears as **"To be scheduled"**, without field or time.	RF-01, HU-01
BR-03	A reported result that the organization has not approved yet is shown and counted in the standings as **"Provisional"**; it becomes **"Final"** only after approval against the paper match sheet.	RF-02, HU-02
BR-04	Standings award 3 points per win, 1 per draw and 0 per loss, and break ties by goal difference, then goals for, then wins. Withdrawn teams go to the bottom, marked **"Withdrawn"**, and all their matches count as 0–3 walkovers.	RF-03, tournament rules
BR-05	Top scorers and best defense accumulate over the whole tournament for each category. Goals from annulled matches do not count.	RF-04
BR-06	The public view **never exposes** players' ID numbers, birth dates, ID-card photos, contact data or balances. A player is shown only by name, team and goals (Colombian Law 1581 of 2012; most players are minors).	RNF-04
BR-07	Only documents explicitly marked as **public** are listed; documents addressed to delegates or referees never appear.	RF-05, HU-05
BR-08	A fan account only grants access to the fan's own data (follows, notifications, orders, bookings). It never grants access to delegate, referee or organizer panels.	RF-16, RNF-05
BR-09	A fan receives a push notification only for teams they follow, and only when that team's schedule is published or its result is approved.	RF-18
BR-10	An order or booking is **confirmed only when the payment gateway approves the payment**. Each gateway transaction is applied **once**, even if the webhook arrives several times (idempotency). The system never stores card data.	RF-23, RNF-06
BR-11	A product variant cannot be sold beyond its stock; stock is reserved when the payment is approved.	RF-22, RF-23
BR-12	A photo/video booking is allowed only for a **scheduled** match and only up to the booking cut-off set by the organizer; a photographer is assigned per match, not per booking.	RF-25, RF-26
BR-13	The customer is notified (e-mail + push) as soon as the photographer uploads the delivery link; the link is visible only to the customer who paid.	RF-27
BR-14	A sponsor placement is shown only between its start and end dates, and sponsors only see the metrics of their own placements. Impressions and clicks are recorded **without personal data** of the visitor.	RF-19, RF-20, RF-21
BR-15	Each organization has its own brand (logo, colors, domain) and its data never mixes with another organization's (multi-tenant isolation by RLS, not by hostname).	RF-13, RNF-11
BR-16	A pre-registration does not create a team in the tournament; the organizer must approve it, and only then is the club invited to the regular registration flow of the management platform.	RF-28

2. CRC cards

One card per main class of the Public Module. Classes marked **read-only** come from the tournament-management platform through the `public_*` views; the Public Module never changes them. Classes marked **owned** are new in this module.

PublicSite *(owned)*

Responsibilities	Collaborators
Hold the organization's public brand: logo, primary and secondary colors, custom domain.	Organization
Decide which tournaments are published and which sections are on (news, gallery, store, bookings, pre-registration).	Tournament
Resolve the organization from the request host.	Organization
Provide the PWA manifest (name, icon, colors).	—

Tournament *(read-only)*

Responsibilities	Collaborators
Know its name, dates, status and whether it is public.	Organization
List its categories, groups and matchdays.	Category, Match
Refuse to expose any data when it is not public.	PublicSite

Team *(read-only)*

Responsibilities	Collaborators
Know its name, crest, club, category and group.	Club, Category
Give its upcoming matches and recent results.	Match
Give its current position in the group.	Standing
Report whether it has withdrawn.	Standing

Club *(read-only)*

Responsibilities	Collaborators
Know its name, slug and logo.	—
List all its teams in a tournament, grouped by category.	Team, Category

Match *(read-only)*

Responsibilities	Collaborators
Know home and away teams, date, time, field and matchday.	Team, Venue/Field

Responsibilities	Collaborators
Expose its public status: to be scheduled, scheduled, provisional, final, postponed, walkover.	—
Give the score and the scorers (name and team only).	Goal
Tell whether it can still be booked for photo/video.	MediaBooking

Standing *(read-only, computed view)*

Responsibilities	Collaborators
Give P, W, D, L, GF, GA, GD and points per team and group.	Team, Match
Order the table with the tie-break rules and send withdrawn teams to the bottom.	Team
Flag rows that include provisional results.	Match

FanAccount *(owned)*

Responsibilities	Collaborators
Register and authenticate a fan (e-mail or Google).	Auth provider
Follow and unfollow teams.	TeamFollow, Team
Keep push subscriptions and notification preferences.	PushSubscription
List the fan's own orders and bookings.	Order, MediaBooking

TeamFollow *(owned)*

Responsibilities	Collaborators
Link one fan with one team (no duplicates).	FanAccount, Team
Tell who must be notified when a team's schedule or result is published.	Notification

Notification *(owned)*

Responsibilities	Collaborators
Create one in-app notification per follower when a schedule is published or a result is approved.	TeamFollow, Match
Send it by Web Push to every active subscription and mark it as sent.	PushSubscription, Push service
Deactivate subscriptions that the push service rejects (410 Gone).	PushSubscription
Mark as read.	FanAccount

NewsPost *(owned)*

Responsibilities	Collaborators
Hold title, body, cover image, tag and status (draft or published).	Organization
Publish and unpublish; list newest first.	PublicSite

Responsibilities	Collaborators
Optionally attach a matchday photo gallery.	GalleryPhoto

SponsorPlacement *(owned)*

Responsibilities	Collaborators
Hold the creative, the target link, the slot (website banner, app banner, matchday graphic) and the start and end dates.	Sponsor
Tell whether it is active today.	—
Record anonymous impressions and clicks.	PlacementEvent
Summarize metrics by period for the sponsor's report.	Sponsor

Product *(owned)*

Responsibilities	Collaborators
Hold name, description, photo, price and active flag.	Organization
Manage its variants (size) and their stock.	ProductVariant
Refuse to sell more than the available stock.	OrderItem

Order *(owned)*

Responsibilities	Collaborators
Build the cart from items and compute the total.	OrderItem, ProductVariant
Start the checkout with the payment gateway.	Payment
Change status: pending → paid → ready for pickup → delivered (or cancelled).	Payment, Organizer

Payment *(owned)*

Responsibilities	Collaborators
Keep the gateway reference, amount and status of one charge.	Payment gateway
Apply the gateway webhook only once (idempotent by gateway reference).	Order, MediaBooking
Confirm the order or booking only when the payment is approved.	Order, MediaBooking

MediaBooking *(owned)*

Responsibilities	Collaborators
Book a photo or video package for a scheduled match.	Match, MediaPackage, FanAccount
Get paid through the gateway.	Payment
Receive the delivery link from the assigned photographer and notify the customer.	PhotographerAssignment, Notification

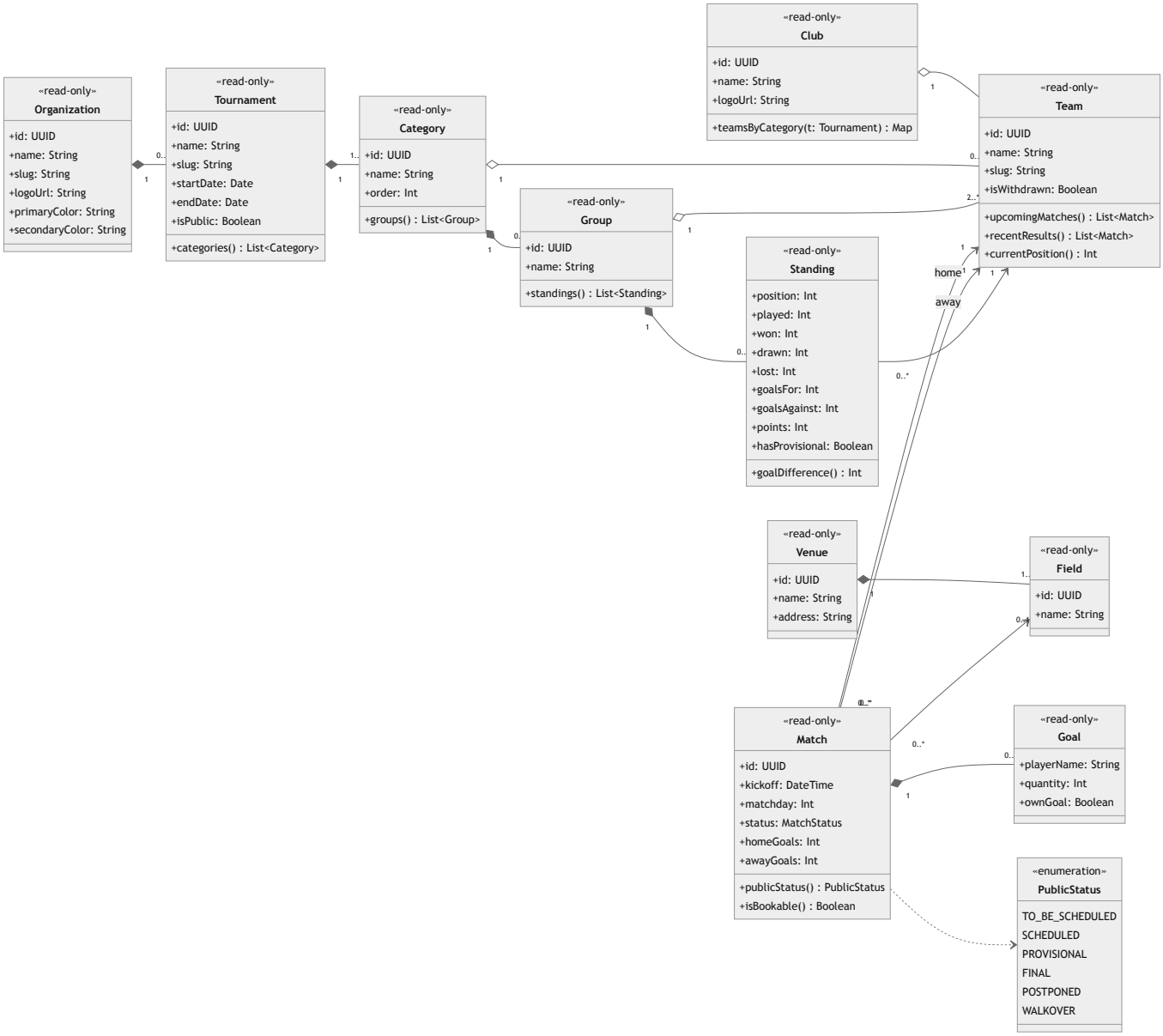
PreRegistration *(owned)*

Responsibilities	Collaborators
Hold the club, contact person, categories of interest and target tournament.	Tournament
Be approved or rejected by the organizer, with a reason.	Organizer
On approval, hand the club over to the regular registration flow of the management platform.	Management platform

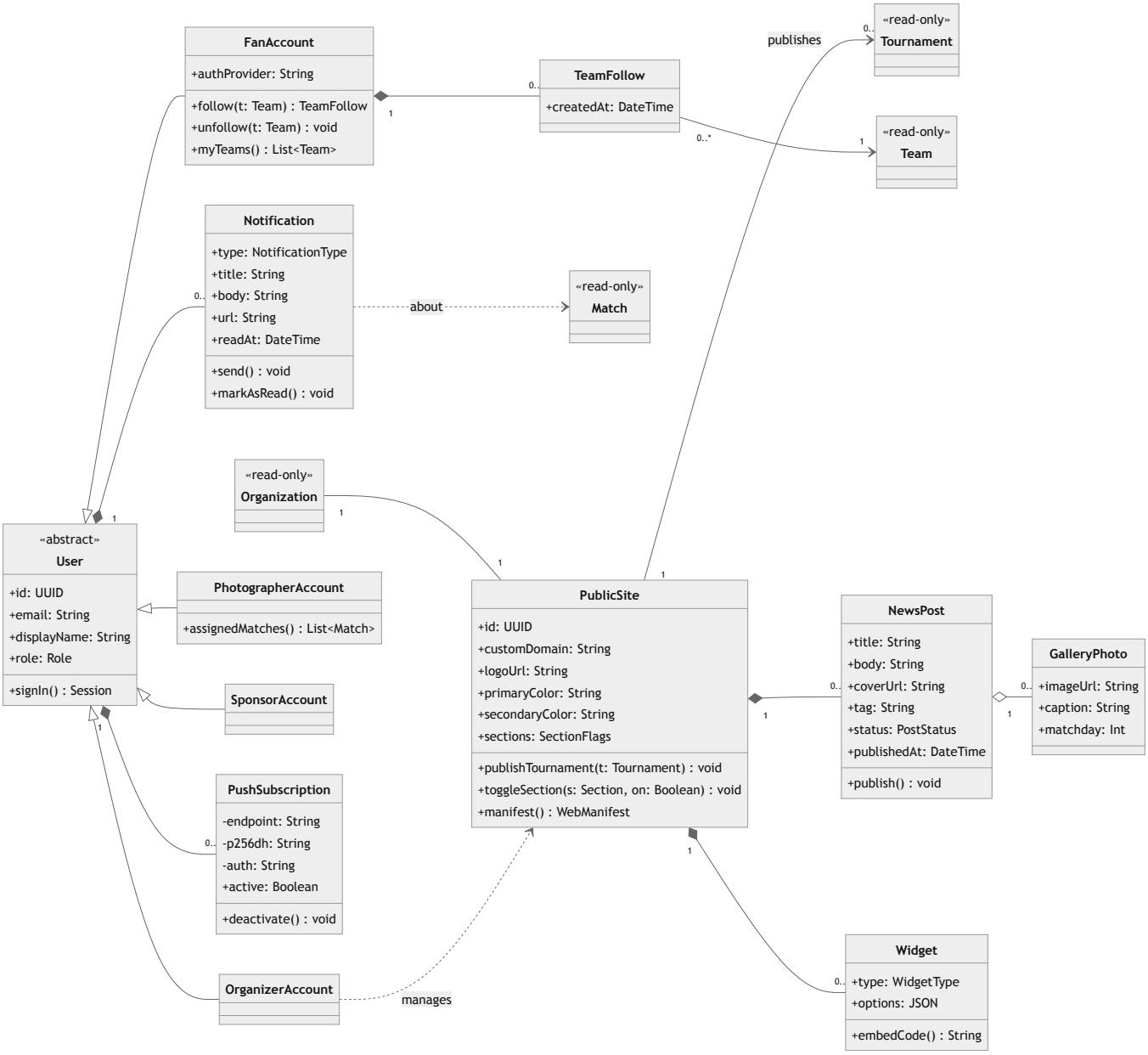
3. Class diagram

UML class diagram of the Public Module, split into three views so it stays readable. Classes with the `«read-only»` stereotype are projections of the tournament-management platform (through the `public_*` views). Every other class is owned by this module. Visibility: `+` public, `-` private.

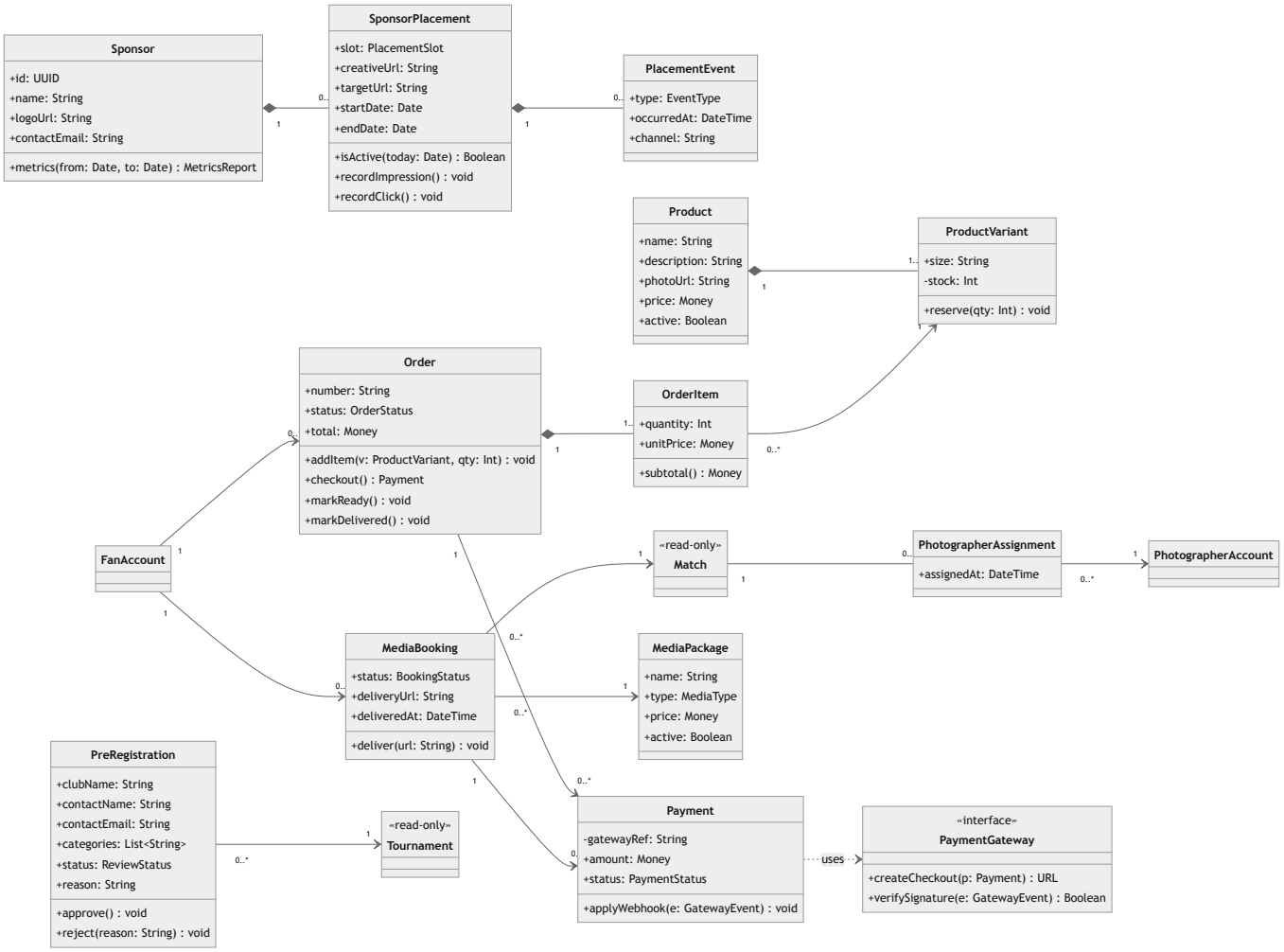
3.1 Competition (read-only projection)



3.2 Public site, fans and content



3.3 Sponsorship and commerce



3.4 Design notes

- **Read-only projection.** Competition classes have no mutating methods in this module: the management platform owns them (Domain-Driven Design *bounded context*). The Public Module talks to them only through views, which keeps the dependency one-way.
- **Dependency inversion.** `Payment` depends on the `PaymentGateway` interface, not on a concrete provider, so the gateway can change without touching orders or bookings (SOLID: DIP and OCP).
- **Single responsibility.** `Notification` only builds and delivers messages; deciding *who* gets them belongs to `TeamFollow`; computing standings stays in the platform's `Standing` view.
- **Roles by specialization.** `FanAccount`, `OrganizerAccount`, `PhotographerAccount` and `SponsorAccount` specialize `User`. In the database this becomes one `pm_accounts` table linked to the authentication provider (`auth.users`) with a `role` column (see section 4).

4. Relational database model

Engine: PostgreSQL (Supabase), the same database as the tournament-management platform.

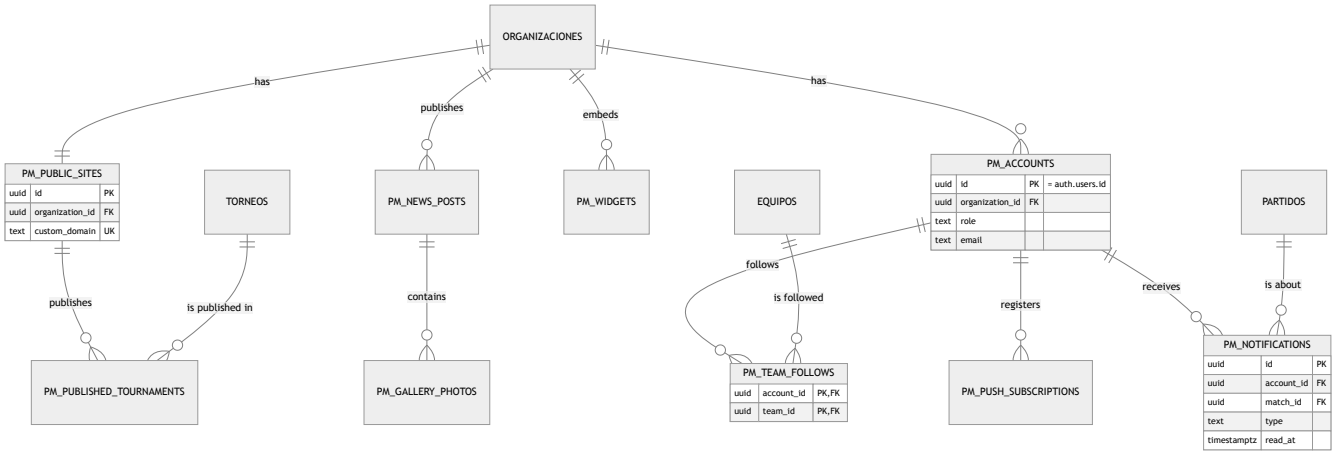
The model has two parts:

1. **Consumed (read-only).** These are the existing platform tables (`organizaciones`, `torneos`, `categorias`, `clubes`, `equipos`, `partidos`, `partido_goles` …). The anonymous role never reads them directly. It reads `public_*` views, which return rows only for tournaments with `torneos.publico = true` and expose no personal data (BR-01, BR-06).
2. **Owned.** These are new tables with the `pm_` prefix (Public Module), created through versioned SQL migrations. Every table is anchored to an organization (`organization_id`), and row-level security (RLS) policies isolate organizations and owners (BR-08, BR-15).

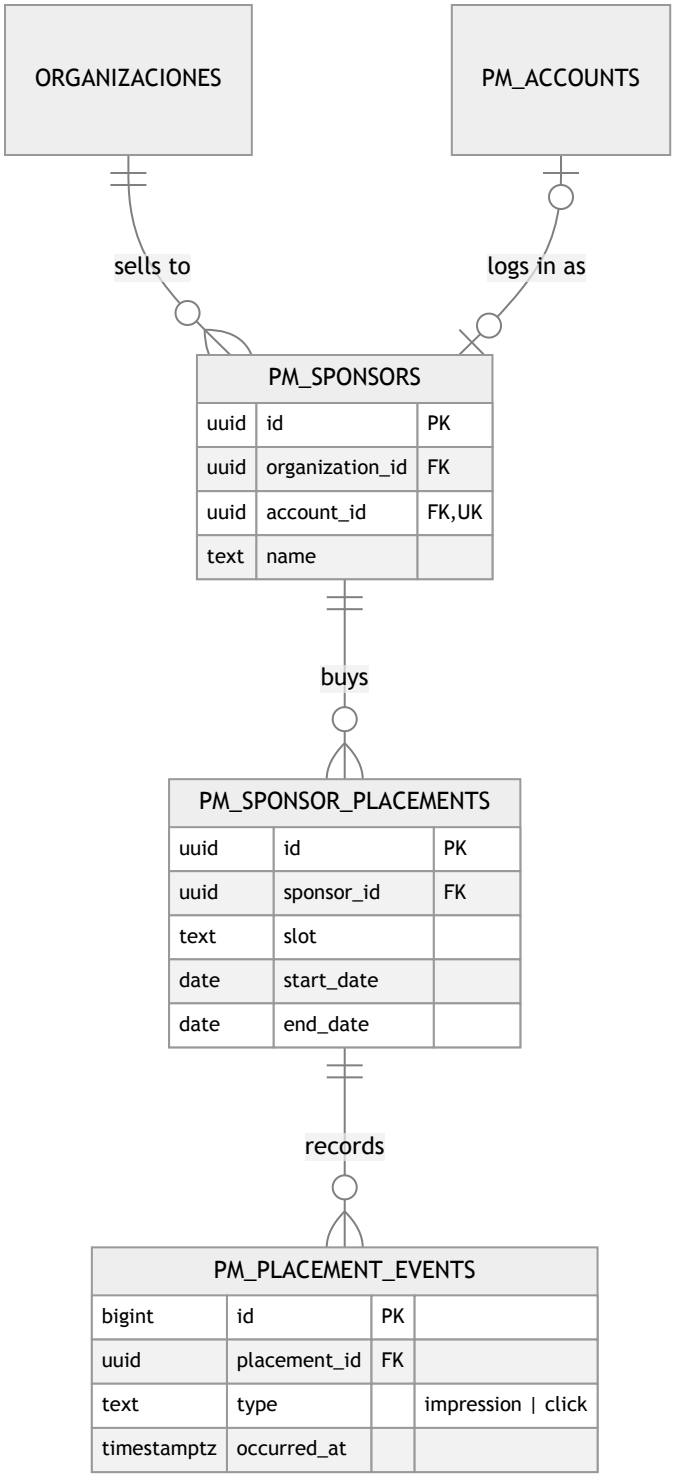
4.1 ER diagrams: owned tables and their links to the platform

Split into three diagrams for readability. Upper-case Spanish names (`ORGANIZACIONES`, `TORNEOS`, `EQUIPOS`, `PARTIDOS`) are existing platform tables, kept as they are in production. Every column of the owned tables is defined in 4.3.

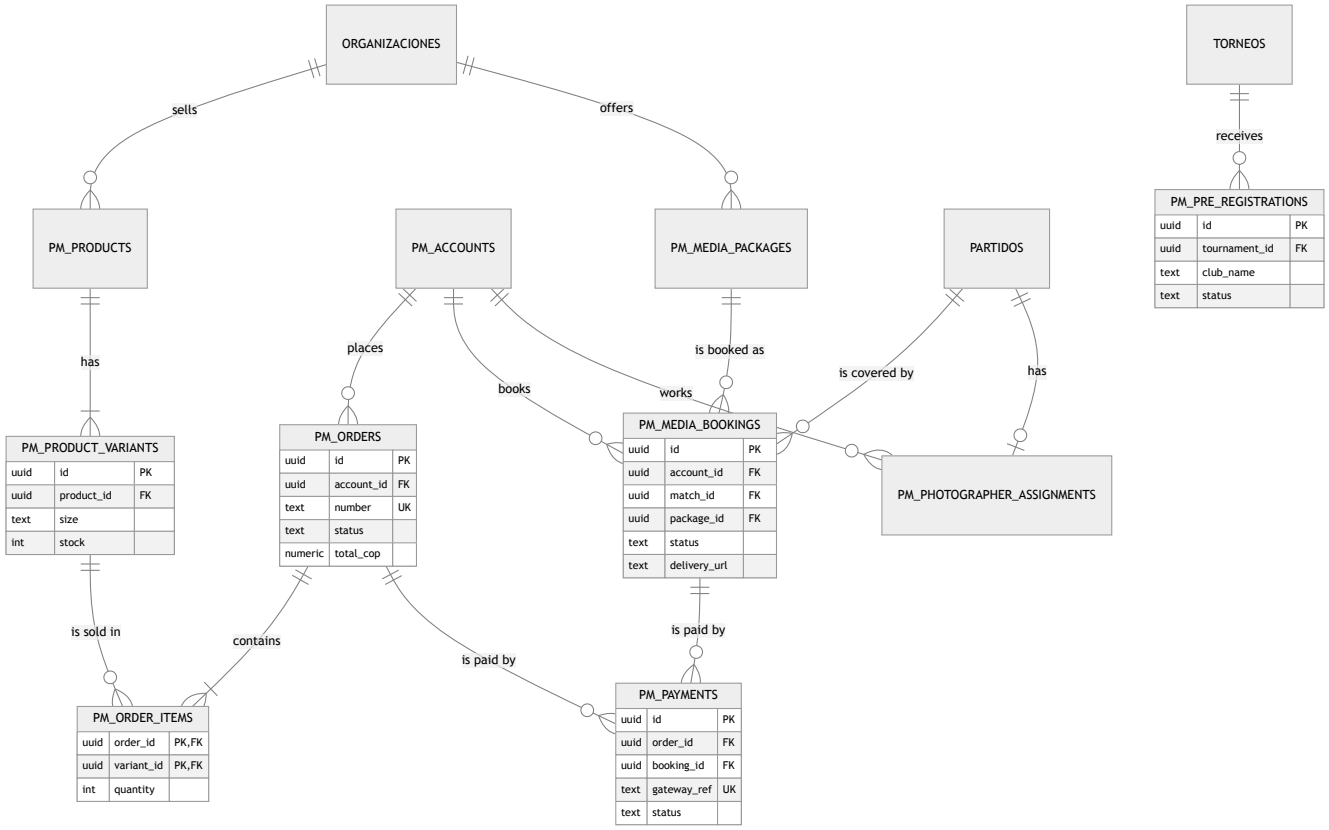
a) Public site, accounts, follows and notifications



b) Sponsorship



c) Store, photo/video bookings, payments and pre-registrations



4.2 Consumed views (read-only, role `anon`)

View	Columns exposed	Source tables	Filter
`public_schedule`	match_id, tournament_id, category, group, matchday, kickoff, venue, field, home_team, away_team, home_goals, away_goals, public_status	`partidos`, `equipos`, `canchas`, `sedes`, `jornadas`	tournament is public; schedule approved, otherwise `public_status = 'TO_BE_SCHEDULED'` and no field or time
`public_match_scorers`	match_id, team_id, player_name, goals	`partido_goles`, `inscripciones`, `participantes`	only the player's name: no ID number, birth date or photo
`public_standings`	tournament_id, category, group, team_id, team, position, played, won, drawn, lost, gf, ga, gd, points, has_provisional, withdrawn	view `posiciones`	tournament is public
`public_top_scorers`	tournament_id, category, player_name, team, goals, position	view `goleadores`	tournament is public
`public_best_defense`	tournament_id, category, team, played, goals_against, position	view `valla_menos_vencida`	tournament is public

View	Columns exposed	Source tables	Filter
`public_teams` / `public_clubs`	id, name, slug, logo_url, category, group, club	`equipos`, `clubes`, `grupo_equipos`	tournament is public
`public_documents`	id, title, type, published_at, file_url	`documentos_torneo`	document marked public (new flag, BR-07)

4.3 Owned tables

Notation: **PK** primary key, **FK** foreign key, **UK** unique. Every table also has `created_at timestamptz DEFAULT now()`.

Public site and content

Table	Columns	Keys and constraints
`pm_public_sites`	id uuid, organization_id uuid, custom_domain text, logo_url text, primary_color text, secondary_color text, show_news bool, show_gallery bool, show_store bool, show_bookings bool, show_preregistration bool	PK id · FK organization_id → organizaciones(id) · UK organization_id · UK custom_domain
`pm_published_tournaments`	site_id uuid, tournament_id uuid, published_at timestamptz	PK (site_id, tournament_id) · FK site_id → pm_public_sites · FK tournament_id → torneos(id)
`pm_news_posts`	id uuid, organization_id uuid, title text, body text, cover_url text, tag text, status text, published_at timestamptz, author_id uuid	PK id · FK organization_id → organizaciones · FK author_id → pm_accounts · CHECK status IN ('draft','published')
`pm_gallery_photos`	id uuid, post_id uuid, image_url text, caption text, matchday int	PK id · FK post_id → pm_news_posts ON DELETE CASCADE
`pm_widgets`	id uuid, organization_id uuid, type text, options jsonb	PK id · FK organization_id → organizaciones · CHECK type IN ('standings','upcoming','results')

Accounts, follows and notifications

Table	Columns	Keys and constraints
`pm_accounts`	id uuid, organization_id uuid, role text, display_name text, email text	PK id (= auth.users.id) · FK organization_id → organizaciones · CHECK role IN ('fan','organizer','photographer','sponsor') · UK (organization_id, email)
`pm_team_follows`	account_id uuid, team_id uuid	PK (account_id, team_id) · FK account_id → pm_accounts ON DELETE CASCADE · FK team_id → equipos(id)

Table	Columns	Keys and constraints
`pm_push_subscriptions`	id uuid, account_id uuid, endpoint text, p256dh text, auth text, active bool, last_ok_at timestamptz	PK id · FK account_id → pm_accounts ON DELETE CASCADE · UK endpoint
`pm_notifications`	id uuid, account_id uuid, match_id uuid, type text, title text, body text, url text, push_sent_at timestamptz, read_at timestamptz	PK id · FK account_id → pm_accounts · FK match_id → partidos(id) · CHECK type IN ('schedule_published','result_approved','delivery_ready')

Sponsorship

Table	Columns	Keys and constraints
`pm_sponsors`	id uuid, organization_id uuid, account_id uuid, name text, logo_url text, contact_email text	PK id · FK organization_id → organizaciones · FK account_id → pm_accounts (nullable) · UK account_id
`pm_sponsor_placements`	id uuid, sponsor_id uuid, slot text, creative_url text, target_url text, start_date date, end_date date	PK id · FK sponsor_id → pm_sponsors ON DELETE CASCADE · CHECK slot IN ('web_banner','app_banner','matchday_graphic') · CHECK end_date >= start_date
`pm_placement_events`	id bigint, placement_id uuid, type text, channel text, occurred_at timestamptz	PK id · FK placement_id → pm_sponsor_placements · CHECK type IN ('impression','click') · no personal data columns (BR-14)

Store, bookings and payments

Table	Columns	Keys and constraints
`pm_products`	id uuid, organization_id uuid, name text, description text, photo_url text, price_cop numeric(12,0), active bool	PK id · FK organization_id → organizaciones · CHECK price_cop > 0
`pm_product_variants`	id uuid, product_id uuid, size text, stock int	PK id · FK product_id → pm_products ON DELETE CASCADE · UK (product_id, size) · CHECK stock >= 0 (BR-11)
`pm_orders`	id uuid, organization_id uuid, account_id uuid, number text, status text, total_cop numeric(12,0)	PK id · FK organization_id, account_id · UK number · CHECK status IN ('pending','paid','ready','delivered','cancelled')

Table	Columns	Keys and constraints
`pm_order_items`	order_id uuid, variant_id uuid, quantity int, unit_price_cop numeric(12,0)	PK (order_id, variant_id) · FK order_id → pm_orders ON DELETE CASCADE · FK variant_id → pm_product_variants · CHECK quantity > 0
`pm_media_packages`	id uuid, organization_id uuid, name text, type text, price_cop numeric(12,0), booking_cutoff_hours int, active bool	PK id · FK organization_id · CHECK type IN ('photo','video')
`pm_media_bookings`	id uuid, account_id uuid, match_id uuid, package_id uuid, status text, delivery_url text, delivered_at timestamptz	PK id · FK account_id → pm_accounts · FK match_id → partidos(id) · FK package_id → pm_media_packages · CHECK status IN ('pending','paid','delivered','cancelled') · trigger: match must be scheduled and before the cut-off (BR-12)
`pm_photographer_assignments`	match_id uuid, photographer_id uuid, assigned_by uuid	PK match_id · FK match_id → partidos(id) · FK photographer_id → pm_accounts (role = photographer)
`pm_payments`	id uuid, order_id uuid, booking_id uuid, gateway_ref text, amount_cop numeric(12,0), status text, raw_event jsonb, confirmed_at timestamptz	PK id · FK order_id → pm_orders, booking_id → pm_media_bookings · **UK gateway_ref** (idempotent webhook, BR-10) · CHECK exactly one of order_id / booking_id is not null · CHECK status IN ('pending','approved','rejected')
`pm_pre_registrations`	id uuid, tournament_id uuid, club_name text, contact_name text, contact_email text, contact_phone text, categories text[], status text, reason text, reviewed_by uuid	PK id · FK tournament_id → torneos(id) · CHECK status IN ('pending','approved','rejected') · CHECK reason IS NOT NULL when rejected

4.4 Integrity and security rules in the database

- **RLS on every `pm_` table.** A fan reads and writes only rows where `account_id = auth.uid()`. An organizer reads and writes only rows of organizations where they are a member (the existing `es_miembro_org()` helper). A sponsor reads only their own placements and events. The anonymous role reads only published news, active placements, active products and the `public_*` views.
- **Idempotent payments.** The `UK gateway_ref` constraint plus `INSERT … ON CONFLICT DO NOTHING` guarantee that repeated webhooks confirm a payment once.
- **Stock.** It is decremented in the same transaction that marks the payment as approved; `CHECK stock >= 0` aborts overselling.
- **No personal data in public views.** The views select explicit columns (never `*`) and run with `security_invoker` to keep the platform's RLS in force.

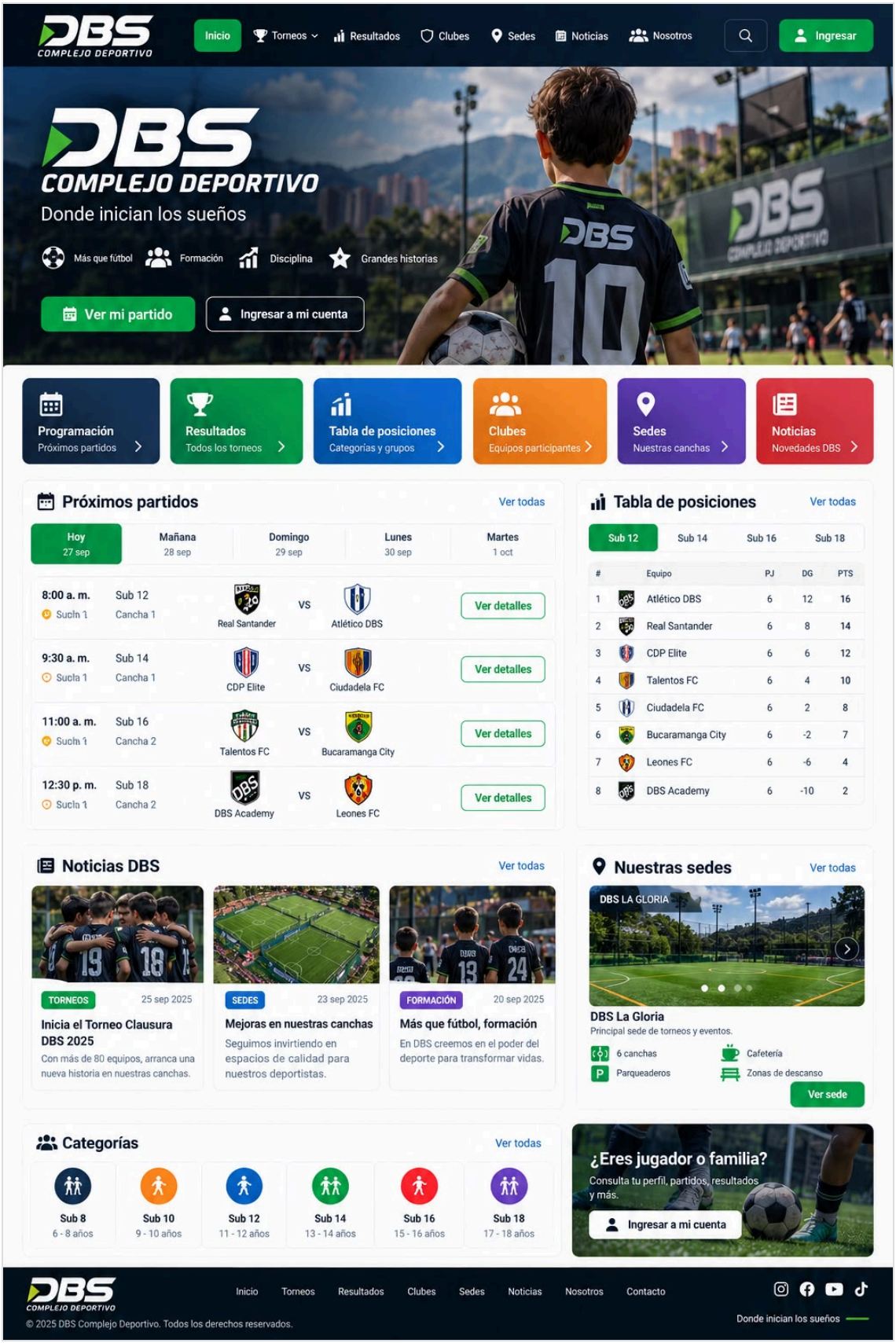
5. Mockups

The mockups cover only the **main screens** of the Public Module. They were built as a clickable high-fidelity prototype in **v0 by Vercel**. The prototype started from a v0 template and used the reference image sent by the pilot organization (DBS Complejo Deportivo) as the visual guide. The exact prompt is in v0-prompt.md.

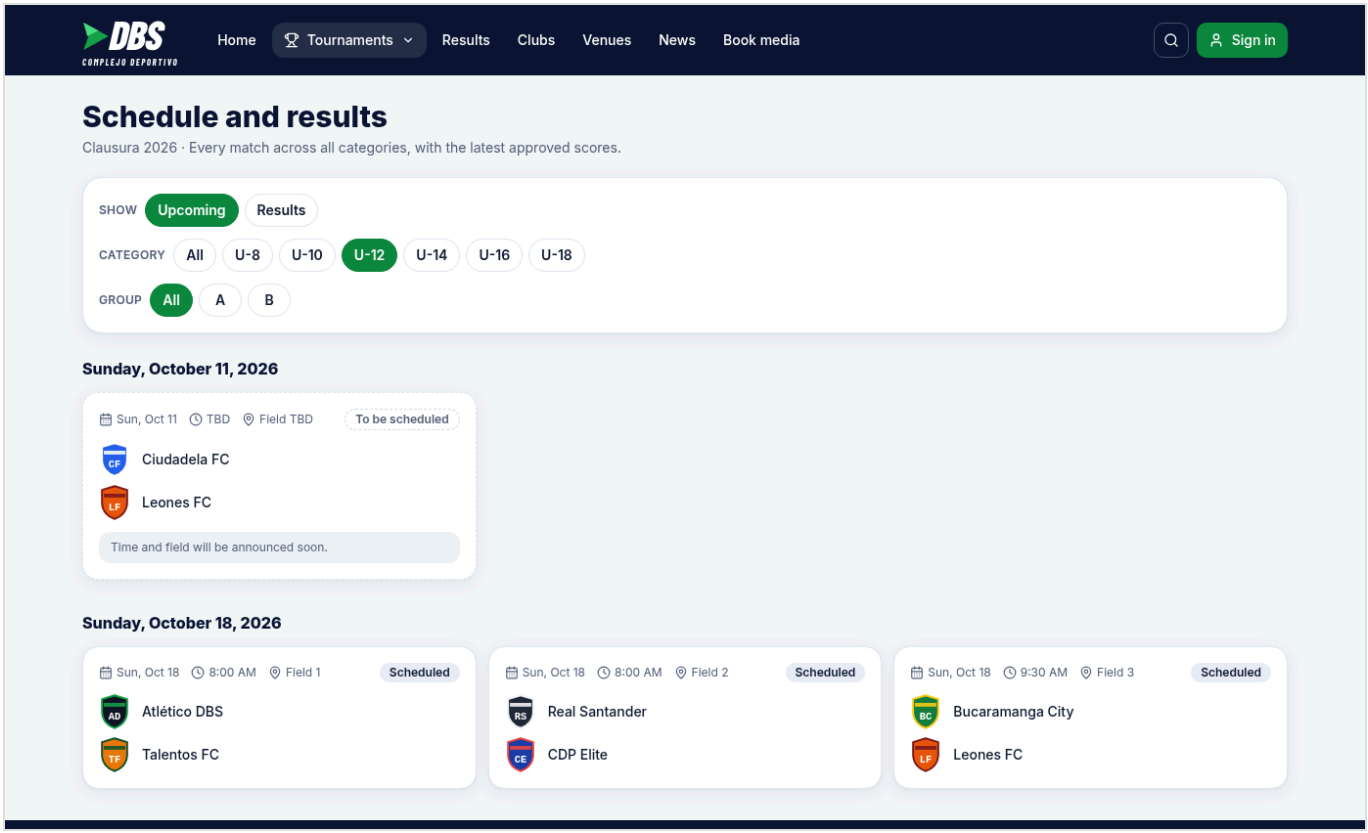
The prototype was requested mobile-first and responsive (RNF-07); the captures below were taken at desktop width. All the data shown is fictional (teams, players, scores and metrics); no real player data is used (BR-06).

#	Screen	Purpose	Requirements covered
0	Reference sent by the organization	Visual identity the public site must follow: dark header, green actions, coloured quick-access tiles, rounded cards. It also defines the home page layout.	—
1	Schedule and results	Lists the matches of the tournament by category and group, with date, time, field and teams. A match without an approved schedule shows **"To be scheduled"** and no time or field (BR-02). The *Results* toggle shows approved scores.	HU-01, HU-02
2	Standings	Group tables with P, W, D, L, GF, GA, GD and points. A yellow dot marks teams with **provisional** results (BR-03) and a withdrawn team goes to the bottom, crossed out (BR-04). Tabs give the top scorers and best defense.	HU-03, HU-04
3	Team profile	One page per team: crest, category, group, position, points, record, upcoming matches and results with scorers (name and team only, BR-06). The **"Follow team"** button starts the fan flow.	HU-07, HU-13
4	Club profile	All the teams of a club in the tournament, one per category, with their group, position and points.	HU-08
5	News	News published by the organization, newest first, filtered by topic.	HU-09
6	Venues	The organization's venue with address, number of fields and amenities, so families know where each match is played.	RF-01 (field of each match)
7	My teams	The fan's home. Without a session it asks the visitor to sign in, because following teams and receiving notifications require a fan account (BR-08).	HU-12, HU-13, HU-14
8	Book photo or video	The fan chooses a package (photo pack or video highlights, prices in COP), picks a scheduled match (BR-12) and requests the booking, which is then paid through the payment gateway (BR-10).	HU-16
9	Sponsorship dashboard	Impressions, clicks and CTR by placement (website, app, match graphics) over the last 60 days, with daily trend. The capture shows the organizer's aggregated view; a sponsor account sees only its own placements (BR-14).	HU-26, HU-33

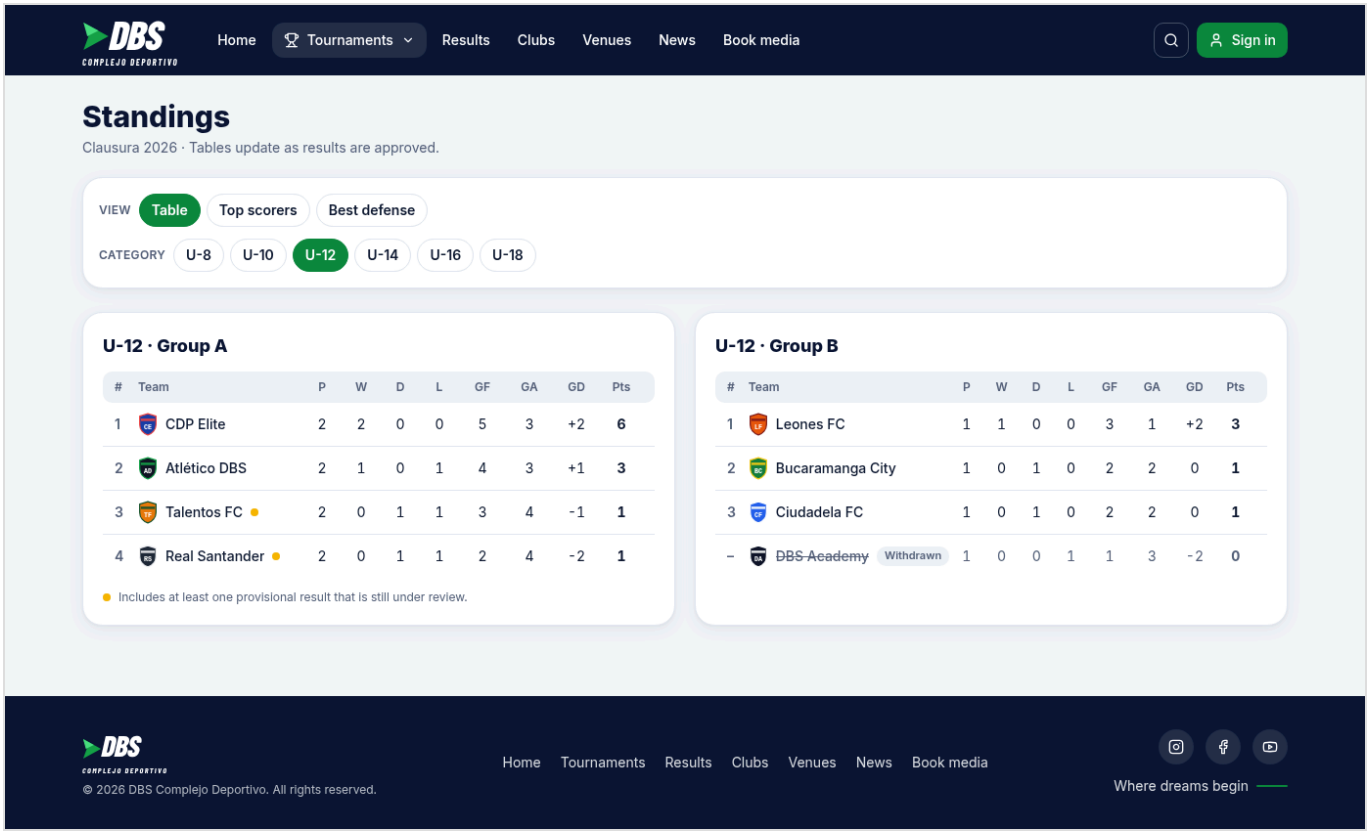
5.0 Reference sent by the organization



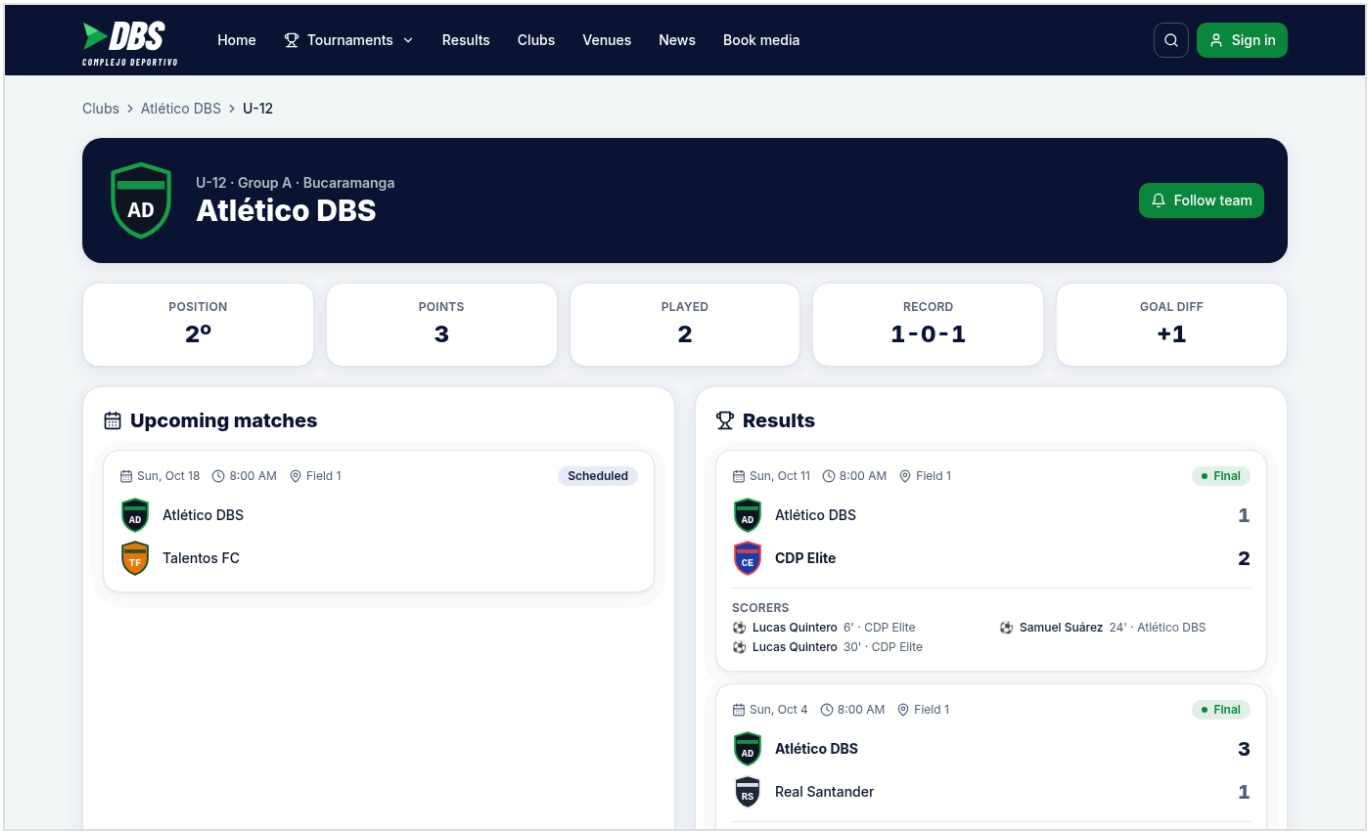
5.1 Schedule and results



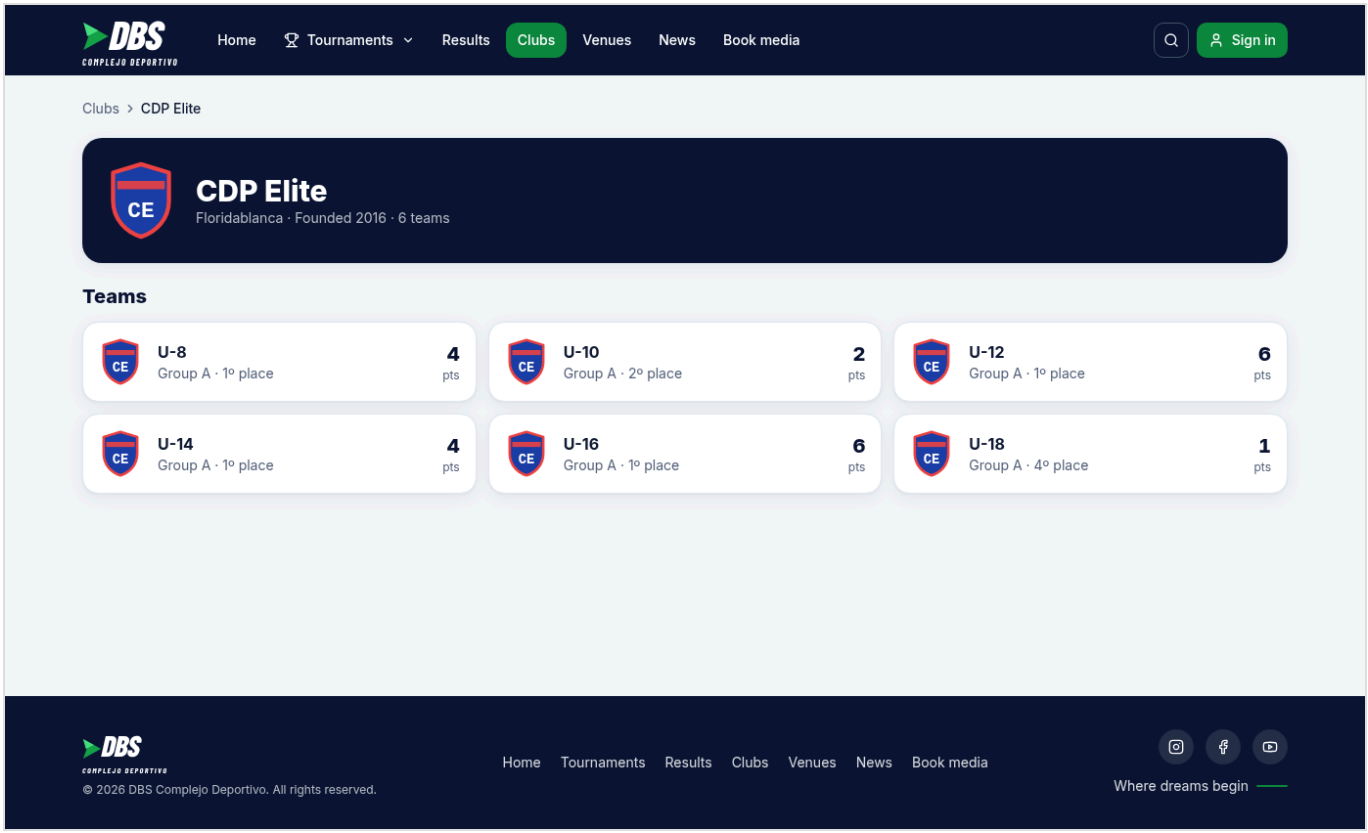
5.2 Standings



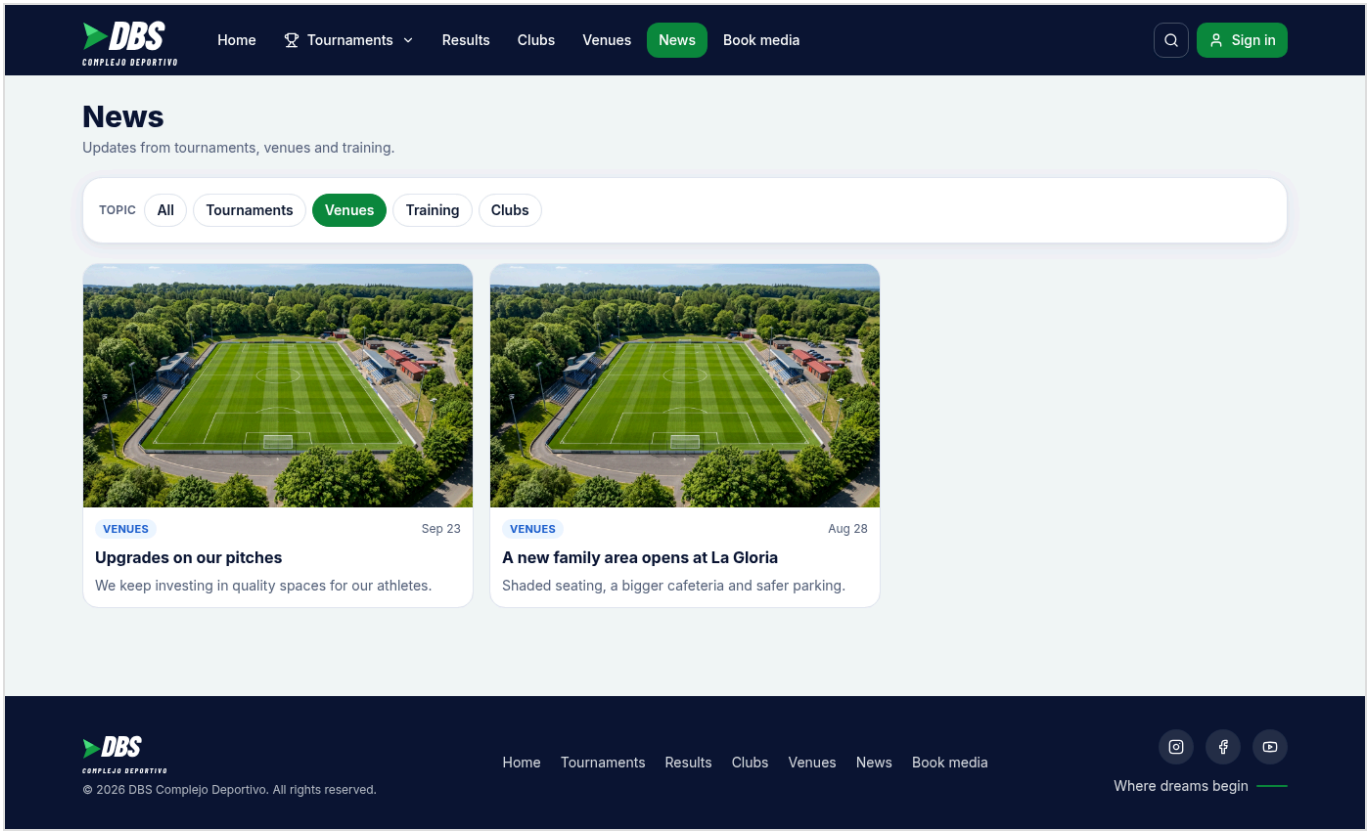
5.3 Team profile



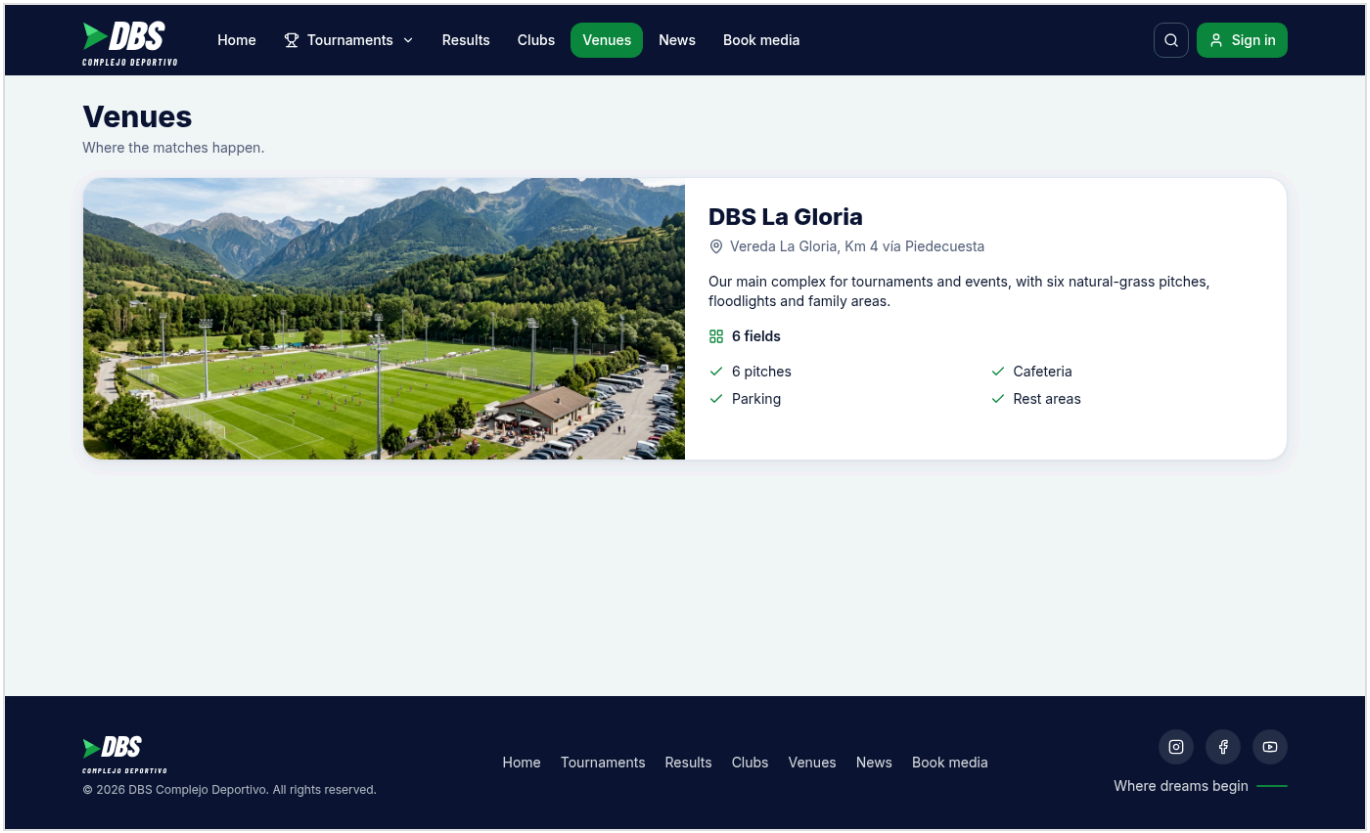
5.4 Club profile


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Clubs > CDP Elite
CE
CDP Elite
Floridablanca · Founded 2016 · 6 teams
Teams
U-8
Group A · 1º place
4
pts
U-10
Group A · 2º place
2
pts
U-12
Group A · 1º place
6
pts
U-14
Group A · 1º place
4
pts
U-16
Group A · 1º place
6
pts
U-18
Group A · 4º place
1
pts
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Where dreams begin

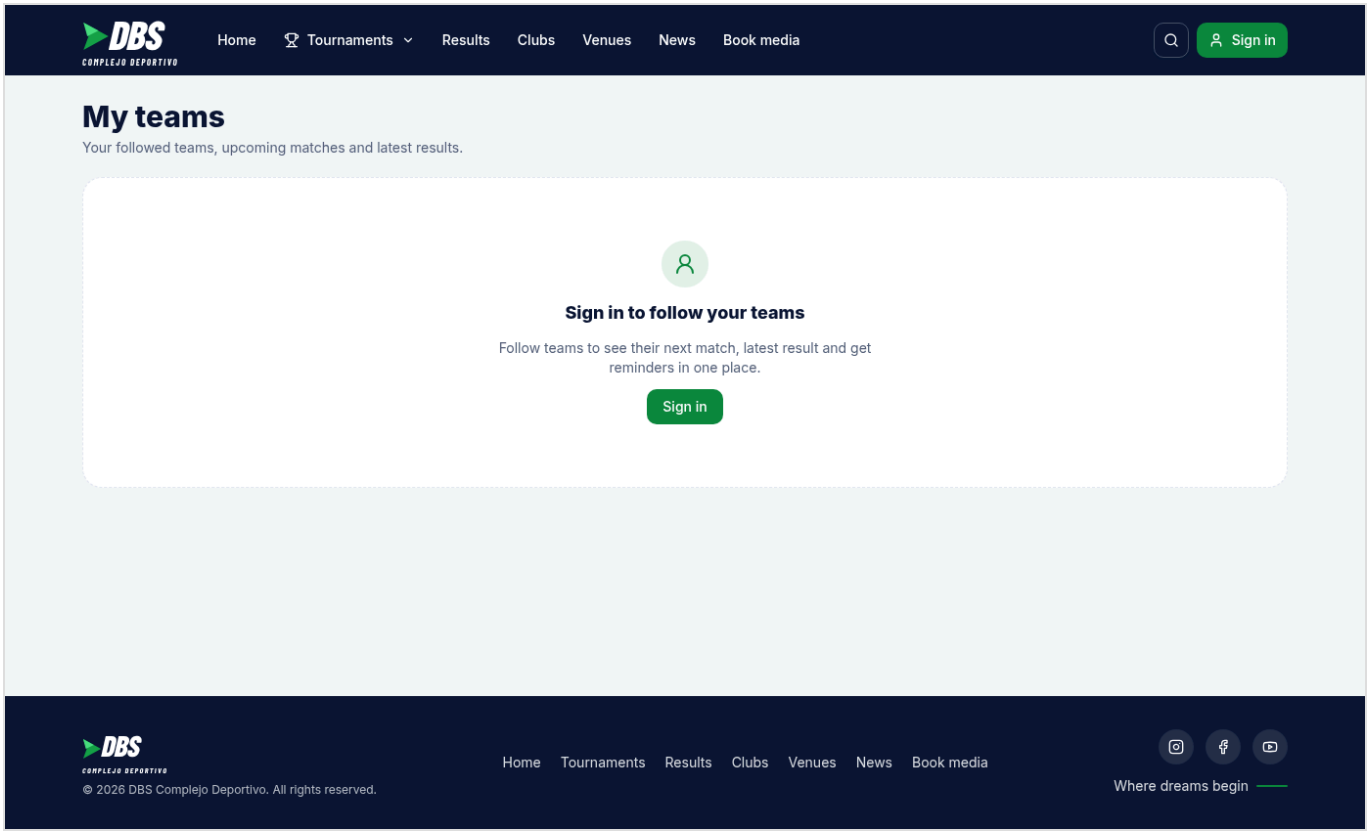
5.5 News


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News
Updates from tournaments, venues and training.
TOPIC
All
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Venues
Training
Clubs
VENUES
Sep 23
Upgrades on our pitches
We keep investing in quality spaces for our athletes.
VENUES
Aug 28
A new family area opens at La Gloria
Shaded seating, a bigger cafeteria and safer parking.
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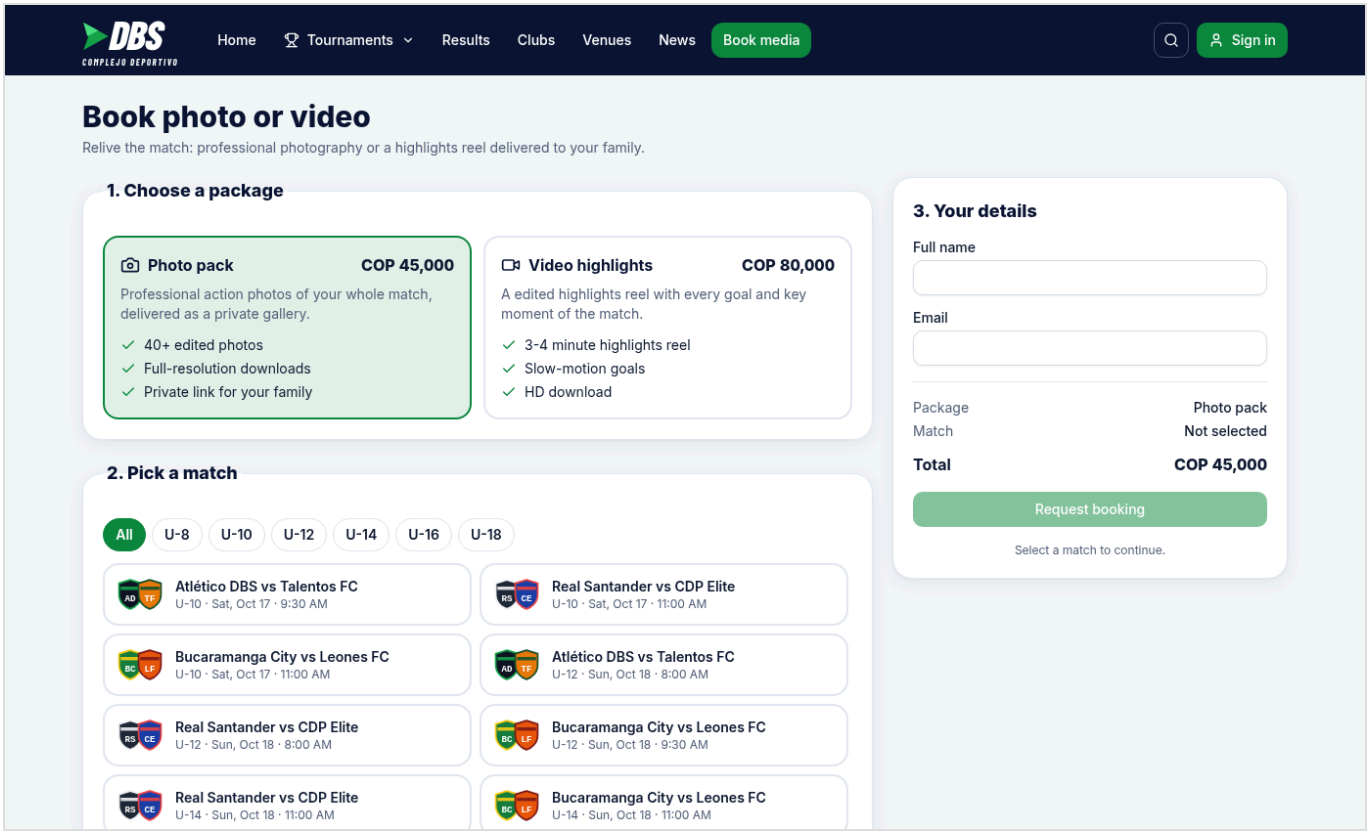
5.6 Venues



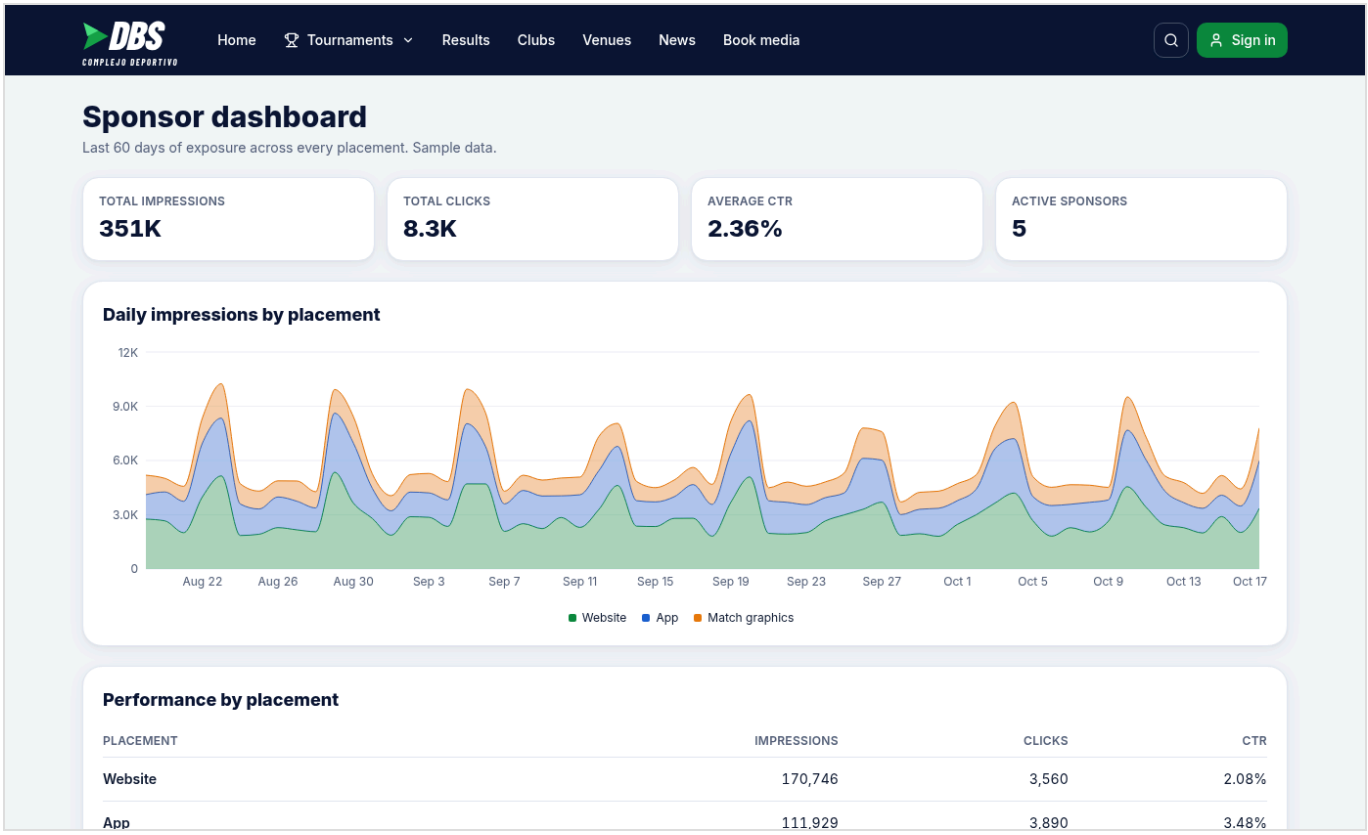
5.7 My teams



5.8 Book photo or video



5.9 Sponsorship dashboard



6. Use of artificial intelligence and references

6.1 Use of artificial intelligence

- **Claude Code (Anthropic, model Claude Opus 5.5).** It drafted the written sections of this workshop: system domain, CRC cards, class diagram, relational model and the mockup prompt. It worked from the Workshop 1 deliverables (requirements, user stories and story map), the approved project proposal and the documentation of the existing tournament-management platform, so that the design reuses the real tables and views. The team reviewed and adjusted every section before submission.
- **v0 by Vercel.** It generated the clickable prototype used for the mockups, from a template, the reference image sent by the organization and the prompt in v0-prompt.md.

6.2 References

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